

VEHICLE CONTROL UNIT (VCU)

Disassembly Manual for Recycling

This manual guides you through the complete disassembly of the VCU into its recyclable fractions. Follow the steps in order. Each step names the components involved and the materials they are made of.

Unit overview

Property	Value
Dimensions	200 × 150 × 60 mm
Weight	approx. 660 g
Tool required	Allen key (hex 2.5 mm)
Estimated time	5 minutes · 5 steps

Components and materials

Component	Qty	Material composition
Housing shells (upper + bottom)	2	Aluminium 363 g · labels & adhesive 15 g
Screws (housing, connectors, board)	14	Steel 15 g · brass threaded inserts (in housing) 4.8 g
Connectors (LIFE, SENS-A, SENS-B)	3	Brass (Cu-Zn) 46 g · polymer insert 11 g · gold plating 0.8 g
Main PCB (circuit board)	1	FR-4 board 91 g · copper 41 g · passive parts 30 g · other (solder, coating) 23 g
Processors	3	Epoxy package · copper leadframe · silicon die (together 19.8 g with chips)
Chips (memory)	3	Epoxy package · copper leadframe · silicon die

Gold-plated connector pins and the silicon components are the highest-value fractions — handle them without damage.

Disassembly steps

Step 1 — Open the housing

- Using the Allen key, remove the 4 housing screws (steel) from the corners of the upper housing.
- Lift the upper housing (aluminium shell) straight up and set it aside.

Materials: screws — steel · upper housing — aluminium.

Step 2 — Remove the connectors

- Remove the 6 connector screws (steel), 2 on each connector.
- Pull each of the 3 connectors (LIFE, SENS-A, SENS-B) straight forward, out of its opening.

Materials: connector bodies — brass with polymer insert · pins — gold-plated (high value).

Step 3 — Lift out the main PCB

- Remove the 4 board screws (steel) holding the circuit board.
- Lift the main PCB straight up, with the processors and chips still mounted on it.

Materials: board — FR-4 (glass-fibre epoxy) with copper layers and passive parts.

Step 4 — Recover the silicon

- Unclip the 3 processors and the 3 memory chips from the board, one by one.
- Place them in a separate container — do not bend or break the packages.

Materials: epoxy package · copper leadframe · silicon die (high value — keep intact).

Step 5 — Sort the housing

- Place the bottom housing together with the upper housing in the metal fraction.
- Remove any labels or adhesive residue if present.

Materials: housing shells — aluminium (largest single material fraction, 363 g).

End of disassembly. All five fractions — aluminium, steel, brass/gold connectors, PCB, and silicon components — should now be separated.

Document v1 · 2026-07-21 · study material